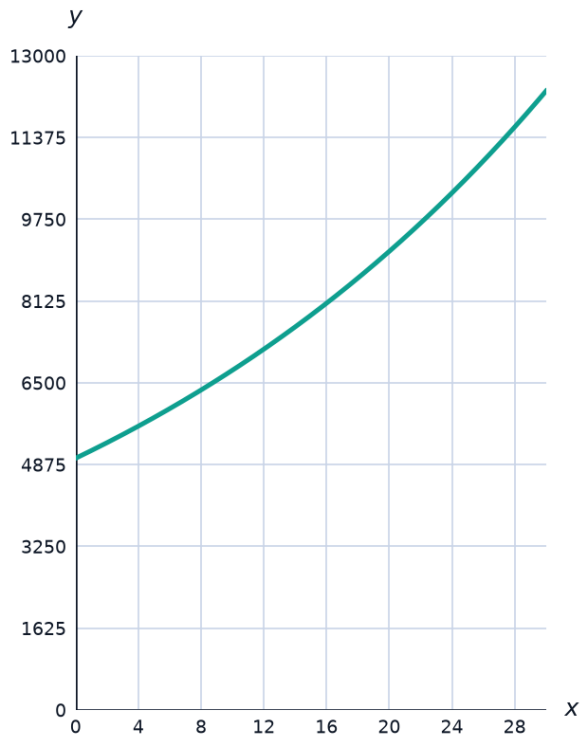
Exponentials & Logarithms

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Exponential growth and decay models**

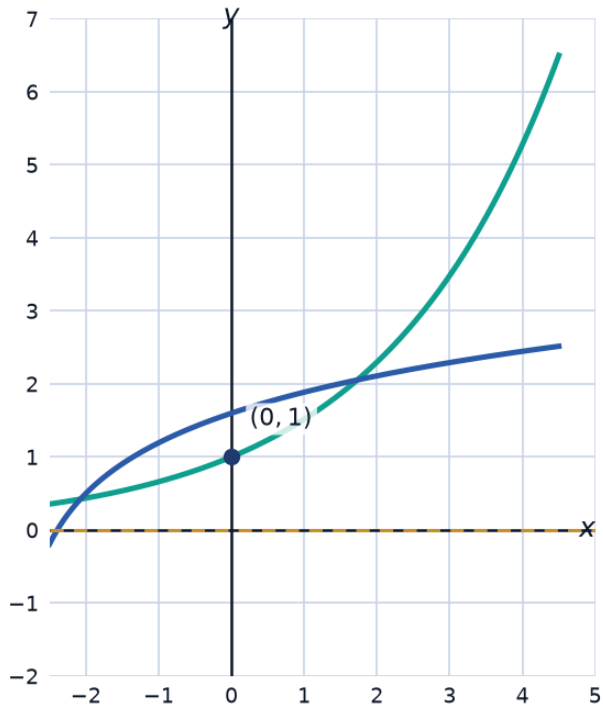
The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.



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2 [Foundational] • Target: Evaluating exponential expressions

Evaluate 2^5.



3 [Foundational] • Target: Evaluating exponential expressions

Evaluate 3^{-2}.

4 [Foundational] • Target: Evaluating exponential expressions

Evaluate $16^{1/2}$.

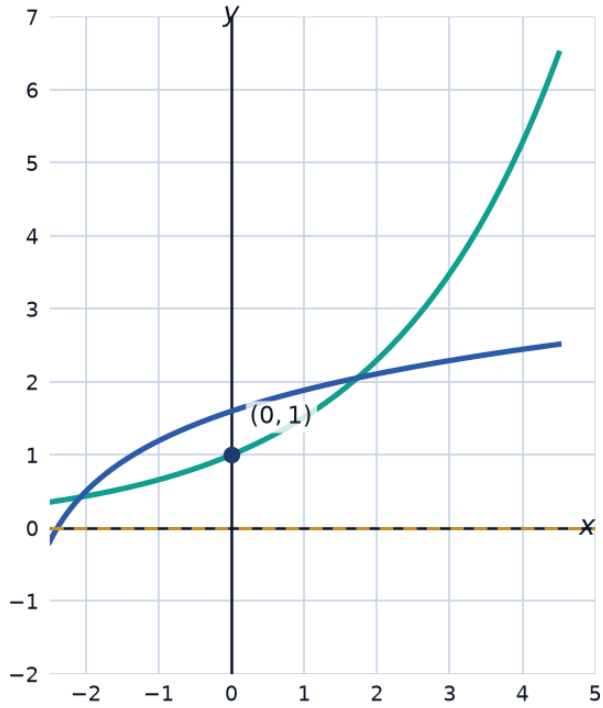
5 [Foundational] • Target: Evaluating exponential expressions

Evaluate $27^{2/3}$.

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6 [Foundational] • Target: Evaluating logarithms

Evaluate $\log_2 32$.



7 [Foundational] • Target: Evaluating logarithms

Evaluate $\log_3 81$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Evaluating logarithms

Evaluate $\log_5 1$.

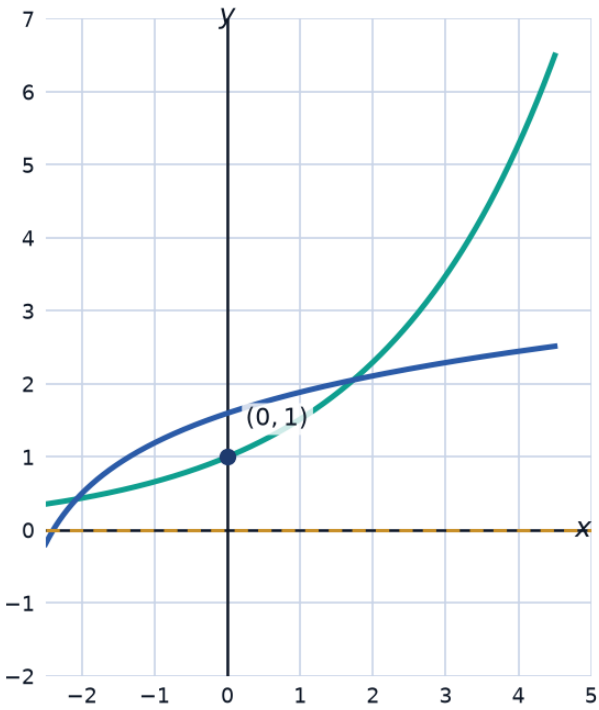
9 [Intermediate] • Target: Evaluating logarithms

Evaluate $\log_{10} 0.01$.

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10 [Intermediate] • Target: Log laws: product and quotient

Write $\log_2(8x)$ as a sum of logarithms.



11 [Intermediate] • Target: Log laws: product and quotient

Write $\log_3\left(\frac{x}{9}\right)$ as a difference of logarithms.

12 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

13 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.

14 [Intermediate] • Target: Log laws: the power rule

Write $\log(x^5)$ using the power rule.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

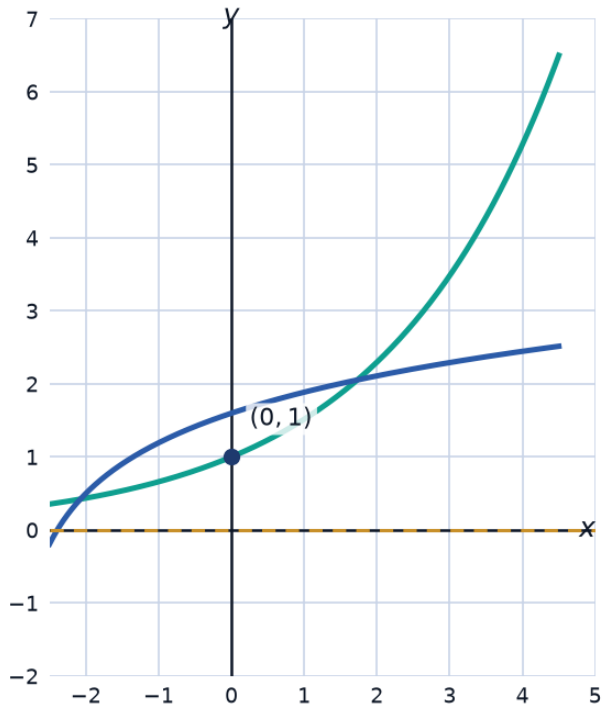
15 [Challenging] • Target: Log laws: the power rule • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\log_2(4^3)$ using the power rule.

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16 [Challenging] • Target: Log laws: the power rule • Cross-Lesson: 7.01 Integration & Applications

Write $2\log_3 x + \log_3 y$ as a single logarithm.



17 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 9.01 Statistics & Probability

Expand $\log\left(\frac{x^3y}{z^2}\right)$ fully.

18 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 1.01 Algebra & Equations

Condense $3\ln x - 2\ln y$ into a single logarithm.

19 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 2.01 Functions & Transformations

Expand $\log_2\left(\sqrt{x} \cdot y^3\right)$ fully.

20 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 3.01 Sequences & Series

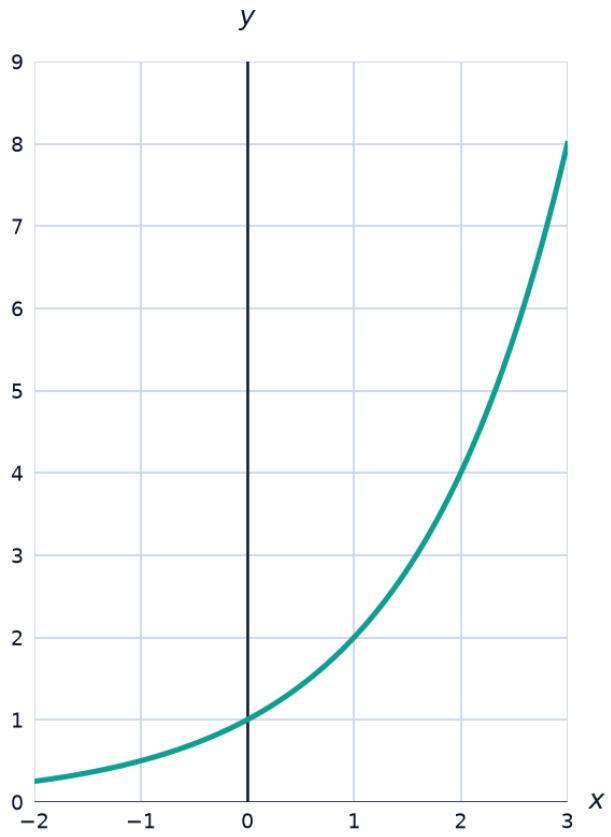
Solve $2^x = 32$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: MCQ — Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?



a. Neither

b. Growth

c. Cannot be determined

d. Decay

22 [Foundational] • Target: MCQ — Evaluating exponential expressions

Evaluate 3^4.

a. 81

b. 243

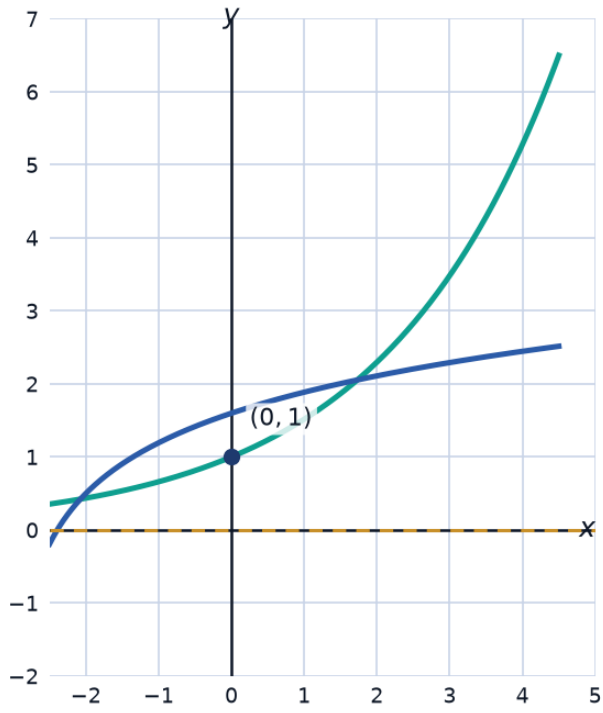
c. 12

d. 64

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23 [Foundational] • Target: MCQ — Evaluating exponential expressions

Evaluate $8^{2/3}$.



a. 5.33

b. 4

c. 2

d. 16

24 [Foundational] • Target: MCQ — Evaluating logarithms

Evaluate $\log_2 64$.

a. 32

b. 8

c. 5

d. 6

25 [Foundational] • Target: MCQ — Evaluating logarithms

Evaluate $\log_5 125$.

a. 5

b. 3

c. 1

d. 25

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26 [Foundational] • Target: MCQ — Log laws: product and quotient

Combine $\log_3 54 - \log_3 2$ into a single logarithm and evaluate.

a. 27

b. 3

c. 26

d. 52

27 [Foundational] • Target: MCQ — Log laws: the power rule

Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$

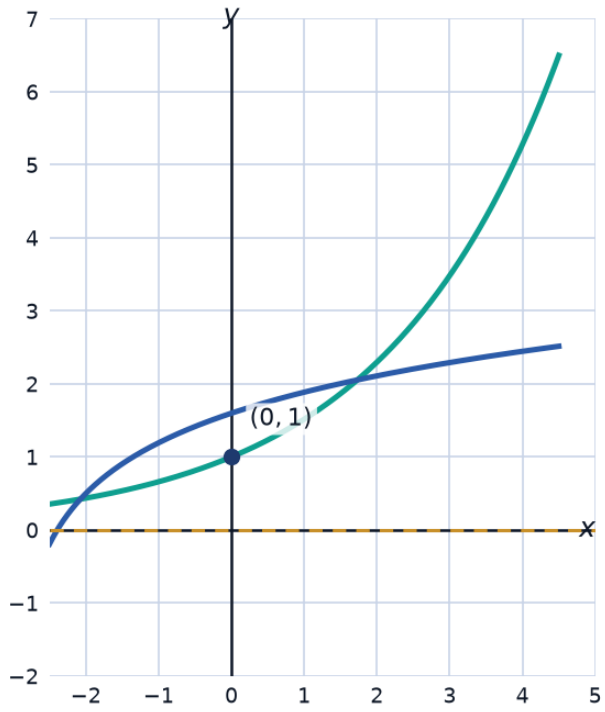
b. $\log(4x - y)$

c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

28 [Foundational] • Target: MCQ — Expanding and condensing logarithmic expressions

Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.



a. $\log(x \cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x} \cdot y^2)$

d. $\log(2\sqrt{x} \cdot y)$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: MCQ — Solving exponential equations (matching bases)**

Solve $5^{3x} = 125$.

a. $x = \frac{1}{3}$

b. $x = 41.67$

c. $x = 3$

d. $x = 1$

30 **[Intermediate]** • **Target: MCQ — Solving exponential equations using logarithms**

Solve $7^x = 50$ for x, rounded to three decimal places.

a. 7.143

b. 2.010

c. 3.000

d. 1.505

31 **[Intermediate]** • **Target: MCQ — Solving logarithmic equations**

Solve $\log_2(x - 1) = 3$.

a. $x = 6$

b. $x = 9$

c. $x = 7$

d. $x = 8$

32 **[Intermediate]** • **Target: MCQ — Natural logarithms and e**

Solve $\ln(x) = 3$ for x, in terms of e.

a. $x = e^{1/3}$

b. $x = 3$

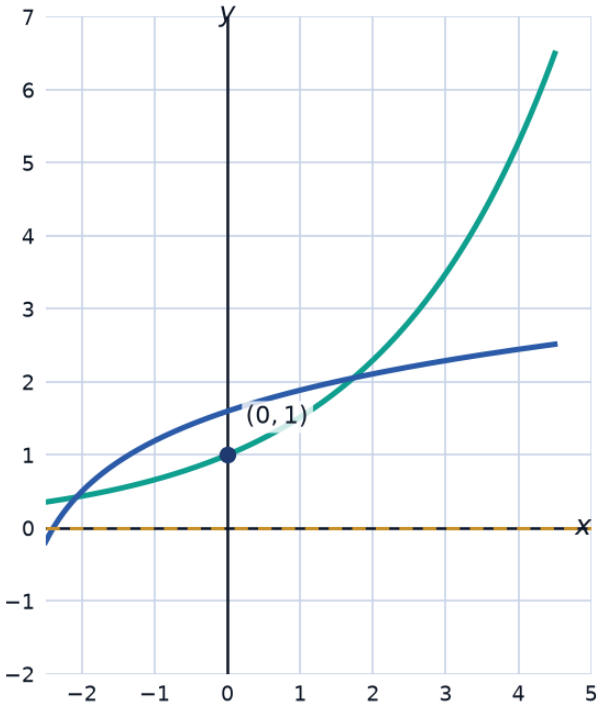
c. $x = e^3$

d. $x = 3e$

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33 [Intermediate] • Target: MCQ — Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.



a. Decay

b. Neither

c. Growth

d. Cannot be determined

34 [Intermediate] • Target: MCQ — Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000

b. 2,000

c. 8,000

d. 1,500

35 [Intermediate] • Target: MCQ — Log laws: the power rule

Evaluate $4^{\frac{1}{2}}$.

a. 4

b. 5

c. 2

d. 6

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36 [Intermediate] • Target: MCQ — Expanding and condensing logarithmic expressions

Evaluate $\log_4 1$.

a. 0

b. 4

c. -1

d. 1

37 [Intermediate] • Target: MCQ — Solving exponential equations (matching bases)

Solve $3^{x+3} = 81$.

a. 4

b. 1

c. 2

d. 0

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Solving exponential equations using logarithms • Cross-Lesson: 1.01 Algebra & Equations

Combine $\log_2 2 + \log_2 2$ into a single logarithm and evaluate.

a. 3

b. 1

c. 4

d. 2

39 [Challenging] • Target: MCQ — Solving logarithmic equations • Cross-Lesson: 2.01 Functions & Transformations

Combine $\log_5 125 + \log_5 25$ into a single logarithm and evaluate.

a. 5

b. 6

c. 3125

d. 4

40 [Challenging] • Target: MCQ — Natural logarithms and e • Cross-Lesson: 3.01 Sequences & Series

Evaluate $\log_3 (27^3)$ using the power rule.

a. 12

b. 9

c. 11

d. 6

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41 [Challenging] • Target: MCQ — Exponential growth and decay models • Cross-Lesson: 4.01 Trigonometry

Solve $\log_4(x-1)=4$.

a. 257

b. 255

c. 256

d. 259

42 [Challenging] • Target: MCQ — Word problems: exponential models in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $3^x = 80$ for x, rounded to three decimal places.

a. 4.090

b. 3.890

c. 3.990

d. 1.903

43 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits & Derivatives

Solve $2^x = 70$ for x, rounded to three decimal places.

a. 1.845

b. 6.130

c. 6.030

d. 6.230

44 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\ln(e^2)$.

a. 1

b. e^2

c. 2

d. 3

45 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Statistics & Probability

A quantity is modeled by $N(t) = 1000e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 545.9

b. 606.5

c. 667.1

d. 1000

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